

THE BONDI DIPOLE IN FULL NUMERICAL RELATIVITY: A SELF-ACCELERATING POSITIVE–NEGATIVE MASS BINARY

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Bondi showed in 1957 that bodies of opposite *active* gravitational mass self-accelerate: the negative chases the positive it repels, and the pair runs off together. We evolve this “Bondi dipole” in 3+1 numerical relativity: two complex scalars share identical Klein–Gordon dynamics; only the phantom enters Einstein’s equations with a minus sign—inertial and passive masses positive, active mass negative. Across thirty-four evolutions, a mass-matched pair released at rest accelerates as a unit: the midpoint moves 3.00 ± 0.01 by $t = 200$, reaches $0.056c$ by $t = 400$ with the rate steady to 2%, holds its separation to 1%, and carries zero total signed momentum to $\lesssim 1\%$. The force is gravity on both of its axes: $a \propto d^{-2.03 \pm 0.01}$ over $d = 8\text{--}20$ and $a \propto M^{0.97 \pm 0.06}$ over a factor 2.5 in mass, with $a d^2$ returning the stars’ own mass at every rung. Swapping the sectors inverts the acceleration to two parts in 10^5 ; doubling the shift damping moves the gauge by 5% and the drift by 0.007%; same-sign control pairs hold their centroids to $\lesssim 8 \times 10^{-4}$ even while merging. The phantom star is, to our knowledge, the first asymptotically flat body of negative ADM mass evolved in numerical relativity; alone it survives to $t = 1000$, relaxing outward, its central amplitude falling 0.63% per 100 time units.

I. INTRODUCTION

Bondi [1] gave negative mass its modern treatment, distinguishing inertial, passive and active gravitational mass and noting the consequence of mixing signs: a negative-mass body is *attracted* by a positive one, the positive body is *repelled* by the negative one, and the pair runs off together (Fig. 1)—momentum conserved, because the negative body’s contribution is negative. Bonnor and Swaminarayan [2] found exact solutions for uniformly accelerated opposite-mass pairs, later generalized in Ref. [3]; Forward [5] showed “in tedious detail” that the self-accelerating pair violates neither momentum nor energy conservation, for equal and unequal mass magnitudes alike, so objections to negative mass must be sought elsewhere [6, 7]. Newtonian N -body work has since carried the same sign rules into cosmological structure formation [42, 43].

Why put this in full numerical relativity? Because nothing in Einstein’s equations forces mass to be positive—positivity is a theorem only once energy conditions are imposed [27, 28]—and modern physics keeps producing sources that strain those conditions: Casimir energies, quantum fields in curved spacetime, phantom dark energy [8]. Every traversable wormhole [9] and warp spacetime [10] needs matter of exactly this kind, and the standing objection to all of them is Bondi’s runaway itself—matter that accelerates for free looks like a perpetual-motion machine [6, 7]. That objection has only ever been examined for point particles and prescribed metrics; the nonlinear evolutions that exist for energy-condition-violating matter—ghost-scalar wormholes in spherical symmetry [15, 30], static

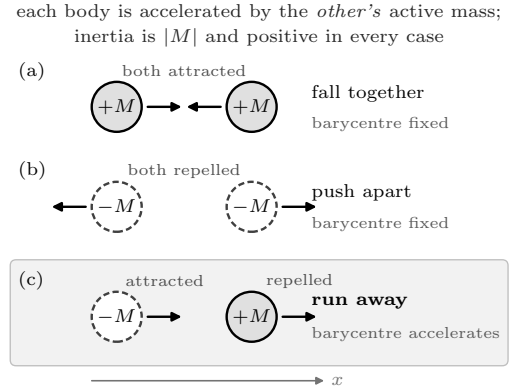


FIG. 1. Bondi’s sign rules for two bodies released at rest, in the style of Forward’s analysis [5]. Every body has positive inertial and passive mass; only the *active* (source) mass carries a sign—negative for the dashed bodies. (a) Two positive-mass stars attract and fall together; (b) two negative-mass stars repel; in both the forces oppose and the pair barycentre stays put. (c) The mixed pair—the Bondi dipole: both forces point the *same* way, so the pair accelerates while the system carries zero total momentum in the signed bookkeeping. Each row is a run family of Table I; in our field-theoretic realization the same-sign rows additionally feel their shared field’s own short-range attraction (Sec. VE), which the mixed row, by construction, cannot.

phantom-field stars [31], and the 3+1 collapse of an Alcubierre warp bubble [11]—concern single objects. Whether the runaway survives full nonlinear gravity—finite bodies, backreaction, radiation—is a checkable question, and we are not aware of a previous 3+1 evolution of a gravitationally bound positive–negative active-mass pair with dynamical, constraint-solved matter.

The question has acquired observational stakes: negative-mass binaries have been proposed as sources

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of anomalous gravitational-wave signatures—anti-chirps, dispersal, runaway motion—turning Bondi’s paradox into an exclusion channel for real detectors [39], while Nojiri and Odintsov [13] find that a bound positive–negative system *can* form for a compact mass immersed in a cosmological fluid with negative cosmological constant. Our asymptotically flat, isolated pair is the complementary case, and it does not bind—it runs away.

The programme: a minimal field-theoretic realization of Bondi’s sign structure (Sec. II); dressed soliton stars of both signs as constraint-satisfying initial data, with the two construction requirements that decide whether the measurement is possible at all (Sec. III); a falsifiable matrix of thirty-four evolutions, each a *cell* of that matrix (Sec. IV); the measurement—a constant-rate, zero-momentum, constant-separation runaway obeying $a = \bar{M}/d^2$, with every control null (Sec. V); the gravitational-wave bound on a doubled box (Sec. VI); and the systematics with their measured sizes (Sec. VII). Every evolution ran on GPUs with `GRTEclyn` [24], the `AMReX`-based [26] GPU port of `GRChombo` [23]; with the worm-hole campaign of Ref. [45] these are, to our knowledge, the first evolutions of energy-condition-violating matter in GPU-native numerical relativity, the warp-bubble collapse of Ref. [11] being the closest CPU-side precedent. Throughout, $G = c = 1$, the scalar mass sets the unit ($m = 1$), X is the pair midpoint (the mean of the two sector barycentres), $\Delta X = X(t) - X(0)$ its drift, and d the instantaneous separation. Runs are named for the sectors they contain, **P** for a canonical star and **M** for a phantom one: **PM** is the mixed pair, **PP** and **MM** the same-sign controls, **MP** the mixed pair with the sectors swapped in place.

II. THE BICOMPLEX SCALAR MODEL

The matter must keep canonical *dynamics*—positive inertial and passive mass—while only its gravitational *source* changes sign; it must form stable, localized stars; and the negative-mass star must be held together by something other than gravity, since its own gravity pushes it apart. Each obvious candidate fails one of these: the negative-mass Schwarzschild spacetime is a naked singularity with exponentially growing smooth linear perturbations [14] and contains no matter from which to build initial data; a negative-energy fluid has no equilibrium to sit in, Bonnor’s static spheres [4] demanding an inverted interior with no non-gravitational restoring force; and a *kinetic-term* ghost—wrong-sign kinetic term, potential untouched—carries its sign flip into the Klein–Gordon equation, where it inverts the effective potential and unbinds the star, and flips inertial and passive mass along with the active one (Forward’s all-negative matter [5], not the signature under test); such fields collapse or explode in evolution [15, 30, 45]. The phantom used here is the other kind of sign flip—an *overall*-sign sector, defined in Eq. (5)—whose field equations are untouched

and whose stress tensor alone is negated.

What passes is a *soliton*: matter self-bound by its own interactions, for which gravity is a dressing rather than the binding agent. We take two independent complex scalars, Φ_+ (*canonical*) and Φ_- (*phantom*), on one spacetime—a *bicomplex* field in the sense used throughout this paper, meaning two independent complex sectors and not a bicomplex-number-valued field—sharing the standard sextic potential of Q-ball and solitonic boson-star studies [16–18],

$$V(|\Phi|^2) = \frac{1}{2}m^2|\Phi|^2 - \frac{1}{4}\lambda|\Phi|^4 + \frac{1}{6}\mu|\Phi|^6, \quad (1)$$

with $\lambda, \mu > 0$: the mass term makes the field oscillate, the attractive quartic lets it clump, the repulsive sextic stops the clumping short of collapse. Sharing one potential makes the matter physics of the two sectors *identical*, so the single sign introduced next is the only possible cause of any asymmetry the simulations show.

That sign lives in one place—the Einstein equations are sourced by the *signed* sum of two canonical stress tensors,

$$G_{\mu\nu} = 8\pi (T_{\mu\nu}[\Phi_+] - T_{\mu\nu}[\Phi_-]), \quad (2)$$

while the field equations of *both* sectors keep their canonical form,

$$\square \Phi_{\pm} = 2 \frac{\partial V}{\partial |\Phi|^2} \Phi_{\pm}, \quad (3)$$

the factor 2 fixed by the $\frac{1}{2}$ -normalized kinetic term. In 3+1 language the sources seen by normal observers accumulate per sector with a sign $\epsilon_{\pm} = \pm 1$,

$$\rho = \sum_{s=\pm} \epsilon_s \rho[\Phi_s], \quad S_i = \sum_{s=\pm} \epsilon_s S_i[\Phi_s], \quad (4)$$

and likewise for S_{ij} , each bracket being the standard scalar expression. The model derives from

$$S = \int d^4x \sqrt{-g} \left[\frac{R}{16\pi} + \mathcal{L}[\Phi_+] - \mathcal{L}[\Phi_-] \right], \quad (5)$$

with $\mathcal{L}[\Phi] = -\frac{1}{2}\nabla_{\mu}\Phi^*\nabla^{\mu}\Phi - V(|\Phi|^2)$: a sector’s overall Lagrangian sign cancels from its Euler–Lagrange equations—hence Eq. (3)—but survives the metric variation—hence Eq. (2). Bianchi consistency is automatic (each sector’s stress tensor is separately conserved on shell, so the signed sum is too); well-posedness is inherited (the sector sign enters CCZ4 only through algebraic source terms); and the mass signature is Bondi’s: the phantom *moves* canonically—positive inertial and passive mass—but *gravitates* with the minus sign of Eq. (2), so its active mass, and its ADM mass, are negative.

The model has exactly two interaction channels, and the run matrix is built to isolate them. The action carries no cross term between Φ_+ and Φ_- —each potential depends on its own $|\Phi_s|^2$ alone—so **two stars of different sectors interact only through the metric**:

gravity is the mixed pair’s only channel, by construction. Two stars of the *same* sector, by contrast, live in one complex field, whose self-interaction gives overlapping lumps a direct, short-range attraction on top of gravity. Section VE measures that field channel—it is strong, and blind to the sign of the mass—and it is precisely why the mixed pair, which cannot have it, is the clean Bondi configuration.

The phantom sector violates the null energy condition by construction, and quantum viability is not claimed—a sector entering gravity with negative energy admits graviton-mediated vacuum decay [32, 33]—so the model is a classical probe of *if such matter existed, what would gravity do*, in the spirit of phantom cosmology [8].

III. INITIAL DATA: DRESSED SOLITONS

A. The stars

A soliton of this theory oscillates coherently, $\Phi \propto e^{-i\omega t}$, and ω labels the solution family: $\omega = m$ is the free-field limit, $1 - \omega$ the binding energy per unit mass. The ratio λ^2/μ sets the lowest attainable frequency, $\omega_{\min} = \sqrt{1 - 3\lambda^2/16\mu}$ [17]; all runs use $\lambda^2/\mu = 4.8$ ($\lambda = 10240$, $\mu \simeq 2.18 \times 10^7$, amplitude $\simeq 0.019$), so $\omega_{\min} = 0.316$.

A flat-space soliton cannot simply be placed on an initial slice: the constraints are violated by a bare Q-ball, and once its own gravity acts the profile is no longer an eigenstate. The star we need is solved with its gravity *on*: the boson-star problem [19–21] with a sextic self-interaction, posed for both source signs. With $\Phi = \varphi_0(r)e^{-i\omega t}$ and a conformally flat, maximally sliced metric, Eqs. (2)–(3) reduce to coupled ODEs for $(\psi, \alpha, \varphi_0)$; **the phantom star is obtained by flipping the sign of the matter sources in the metric equations only**, the Klein–Gordon line keeping its canonical form. The result is a self-interaction-bound soliton with repulsive gravity: $\psi < 1$, $\alpha > 1$ in the core, and negative ADM mass. We shoot at fixed frequency, bisecting the central amplitude and outer-iterating until ω/α_∞ matches the request to $\sim 10^{-5}$.

Figure 2 shows both families. The campaign pair takes the canonical star at $\omega = 0.750$, $M_{\text{ADM}} = +0.014350$, and the phantom at $\omega = 0.7603$, $M_{\text{ADM}} = -0.014295$: mass-matched to 0.39%, because Bondi’s point-mass limit is governed by each body’s exterior field, hence its ADM mass, alone. The pair mean $\bar{M} \equiv \frac{1}{2}(M_+ + |M_-|) = 0.014322$ is what the midpoint acceleration measures. The canonical star has rms radius 4.34 and compactness $M/R \sim 3 \times 10^{-3}$ —two orders below what any horizon needs—and a four-point survey ($\omega = 0.75\text{--}0.90$, $t = 120$) confirms the family static where it is used: the minimum lapse moves by at most 1.2×10^{-3} from birth. Each star is seeded from its own profile table with its own frequency, and the conjugate momentum carries the solved lapse, $\Pi = -(\omega/\alpha(r))\varphi_0(r)$, so the harmonic phase starts on its eigenstate.

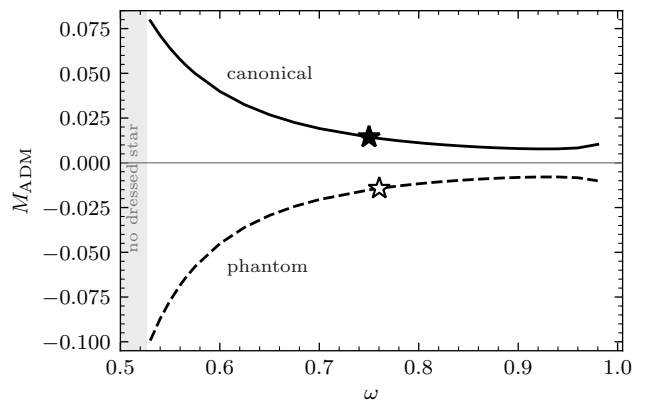


FIG. 2. ADM masses of the spherical stars against oscillation frequency at the working couplings. Lower ω means deeper binding and larger $|M_{\text{ADM}}|$; at equal frequency the phantom star is the heavier in magnitude, its repulsive self-gravity spreading it over more volume. Stars: the campaign pair, mass-matched to 0.4%. Both branches floor at $\simeq 0.54$ of the campaign mass near $\omega \simeq 0.93$, and neither has a bound star at $\omega = 1$ —the bound that shapes the mass ladder of Sec. VF.

B. Two construction requirements

GRTresna [25] then re-solves the constraints for the pair (CTTK family [35], sources summed with their signs star by star). Two choices in this step decide whether the runaway is measurable at all. Both were established by direct testing.

Both sectors must be built on the same slicing. The CTTK algebraic ansatz ties the initial K to the local energy density, $K \propto \sqrt{\rho}$ —imaginary for $\rho < 0$, so a phantom sector forces the maximal-slicing ($K = 0$) path. Letting the canonical sector default to the algebraic path builds the two stars by *different methods*: the canonical star is then born inside a coherent contraction ($\max|K| \approx 0.1$ at $t = 0$, four orders of magnitude above the phantom’s) and squeezed through its first two field periods. We therefore construct *both* sectors with $K = 0$, which for this matter also converges better; the birth kick vanishes ($\max|K| \rightarrow 10^{-5}$, landing on the phantom’s value) and the two constructions differ in nothing but the sign of ρ .

The solve grid must share the evolution grid’s grid-cell centres. The elliptic solve runs on its own (wider) box; if its spacing differs from the evolution’s, the solved metric lands on the evolution grid displaced by a fraction of a grid cell while the matter is repainted exactly—so every star is born off the centre of its own well, the canonical falling back down it and the phantom pushed up it, a directional artefact that does not converge away and was comparable to the signal. Matching the spacings makes the transfer a straight copy: the solved metric moves grid cell by grid cell (a piecewise-constant transfer, exact once the grid-cell centres coincide) while the matter is repainted analytically from its radial profile table. The

metric-minus-matter centroid offset is then 0.0000 on every cell here, and the lone-star drift below is $269\times$ smaller than on mismatched grids. All solves run at the evolution spacing on a doubled box ($L = 128$); exit residuals, tightened with the grid, are in Appendix A.

Every pair places its two lumps at azimuthal angles 0 and π about the box centre—diametrically opposite points $x = \pm d_0/2$ on the pair axis—and paints each from its radial table with the same internal $U(1)$ phase, so all pairs here are seeded *in phase*, relative phase $\delta = 0$. For the mixed pair δ is pure gauge, the two sectors being independent fields with no cross term. For the same-sector controls it is not: two Q-balls of one field attract through the shared field at $\delta = 0$ and repel at $\delta = \pi$ [40, 41], so the seeding used here is the attracting configuration, consistent with the merger those cells show (Sec. V E). Superposed same-field data also carries known phase-dependent constraint systematics that the re-solve reduces but does not eliminate [34]—one reason those cells serve as momentum nulls and not as precision force measurements.

IV. THE CAMPAIGN

A. Numerics

Evolutions use `GRTEClyn`: CCZ4 [22] with damping $(\kappa_1, \kappa_2, \kappa_3) = (3, 0, 1)$, 1+log slicing [36], a Gamma-driver shift [37, 38] with $\eta = 1$, fourth-order stencils, RK4, Kreiss–Oliger dissipation $\sigma = 2$, Sommerfeld outer boundaries, and—because massive-field radiation reflects from massless-wave boundaries—an absorbing sponge layer over $r = 24\text{--}32$ (moved to $48\text{--}60$ on the doubled box). The sponge works: the net inward flux through the accounting boundary stays at round-off ($\lesssim 10^{-7}$) even in the violent same-sign cells. The time step is $\Delta t = 0.02 \Delta x$, an order of magnitude below the usual CCZ4 Courant factor and the dominant cost of the campaign (Appendix B); it is not conservatism but the sextic self-interaction’s own stability limit—at $\Delta t = 0.2 \Delta x$ the star disperses outright.

Every cell is *strictly uniform-grid* (`max_level` = 0), so the resolution ladder means exactly one thing—the grid spacing—and the aligned-grid handoff of Sec. III B is preserved verbatim. The main box is $L = 64$ at $N = 128, 192, 256$ ($\Delta x = 0.50, 0.33, 0.25$ —about 9, 13 and 17 points across a stellar rms radius of 4.3); the wave-zone box doubles L at fixed $\Delta x = 0.25$ with extraction shells at $R = 16\text{--}40$. The reference separation is $d_0 = 10$, about twice the 90%-weight radius, and the separation scan spans $d_0 = 8\text{--}20$. Stars are released *at rest*—no boosts, no driving—so any drift is dynamical. All cells run to $t = 200$ (the stability survey to 120; the long pair run to 400, the lone-phantom control to 1000, and the two late-time follow-up cells of Sec. VII to 784 and 600), and every fitted acceleration a is twice the quadratic coefficient of $X(t)$ over $5 \leq t \leq 200$, the first five units

excluded while the gauge settles; Sec. VII quantifies the (percent-level) sensitivity to that convention.

B. Diagnostics and the run matrix

Each sector is tracked two ways, continuously: a whole-domain barycentre with weight $w_s = (|\Phi_s|^2 + |\Pi_s|^2)^{1/2}$, and a sub-cell core tracker at the sector’s field peak. The headline quantity is the *pair midpoint* $X = \frac{1}{2}(\bar{x}^{(+)} + \bar{x}^{(-)})$: it is what Bondi’s argument predicts to accelerate, it is defined identically for pairs and controls, and by symmetry it is exactly zero for every control—so the control rows of Table I measure the noise floor rather than assume one. The two trackers agree on the pair drift to 1%. Volume-integrated sector momenta, constraint norms, confinement, the $\ell = 2$ modes of Ψ_4 , and energy-condition monitors are recorded alongside. The last return what the model prescribes and nothing more: the minimum NEC is -2.20×10^{-4} in every cell containing a phantom—the lone phantom’s own value, flat over the run—while the lone canonical star returns $+7 \times 10^{-42}$, no violation at all.

Table I is the experiment: each row is a falsifiable prediction of the sign structure, and a control drifting at the rate of the mixed pair would invalidate the physical origin of the runaway.

V. RESULTS

A. The runaway: constant rate, zero momentum, constant separation

Released at rest with the canonical star at $x = 37$ and the phantom at $x = 27$, **both stars accelerate toward $+x$ as a unit** (Figs. 3 and 4). On the finest grid the midpoint moves $\Delta X = +3.0016$ by $t = 200$ at fitted acceleration $a = 1.549 \times 10^{-4}$; a single parabola fits the whole trajectory. The $t = 400$ run (base grid, where $a = 1.45 \times 10^{-4}$) shows the rate is *sustained*: fitted over [133, 266], [300, 400] and [350, 400] the acceleration is 1.451, 1.422 and 1.417×10^{-4} —steady to 2% across a factor three in elapsed time, with the pair at $0.056c$ and still gaining speed linearly when the run ends. There is no impulse, no saturation, and no secondary acceleration.

The two Bondi signatures frame that motion. **Displacement without momentum** (Fig. 4d): each sector’s volume-integrated momentum grows steadily—both reach 5.3×10^{-4} by $t = 200$, the scale set by $M|v|$ —while the *signed* total of Eq. (4) cancels to 3.8×10^{-6} , under 1% of either term, and that residual *converges toward zero* with the grid (3.7×10^{-5} at $N = 128$, 3.8×10^{-6} at $N = 256$), so it is a discretization residual and not a floor. (The cancelling quantity is the matter momentum of Eq. (4), not the ADM momentum—a distinction without numerical consequence at compactness $M/R \sim 3 \times 10^{-3}$.) The pair moves three units while

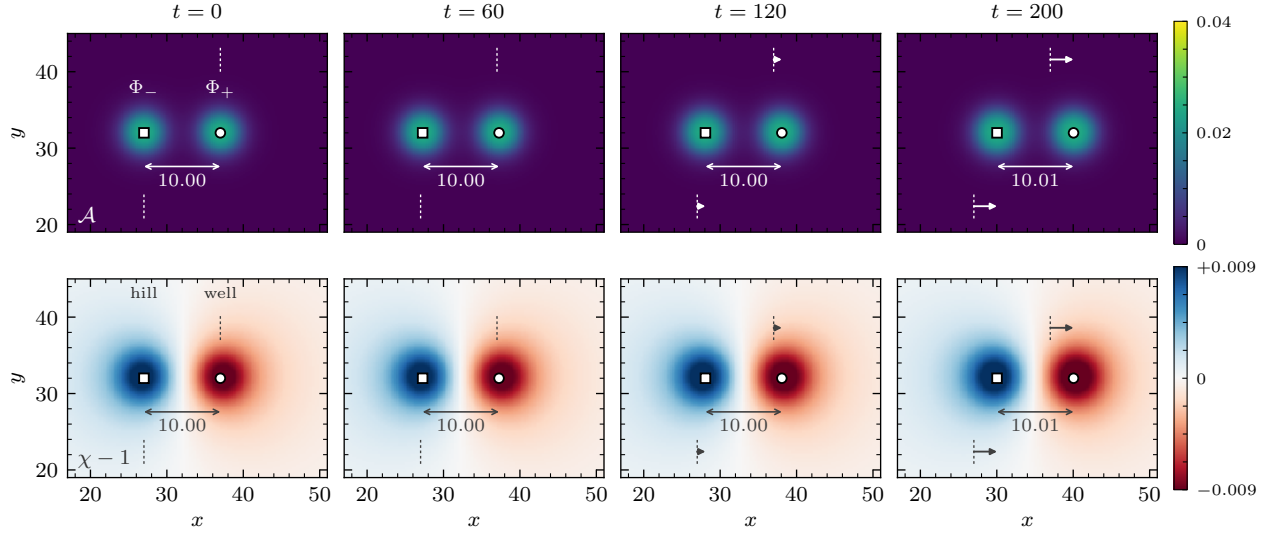


FIG. 3. The runaway in the headline cell (PM, $d_0 = 10$, $\Delta x = 0.25$), $z = 0$ plane, at $t = 0, 60, 120, 200$. Top: scalar-field activity $\mathcal{A} = \sum_s (|\Phi_s|^2 + |\Pi_s|^2)^{1/2}$, both sectors, projected. Bottom: $\chi - 1$; the phantom star is a *hill* ($\chi > 1$, blue, left) and the canonical star a *well* ($\chi < 1$, red, right)—the model’s one sign made visible. Colour bars are locked across stills, symmetric about $\chi - 1 = 0$. Open markers: the tracked cores (circle Φ_+ , square Φ_-); dashed ticks mark the release positions, arrows the displacement since release, and the bracketed number the measured gap. Both stars move toward $+x$ at constant acceleration while the gap holds to 1%.

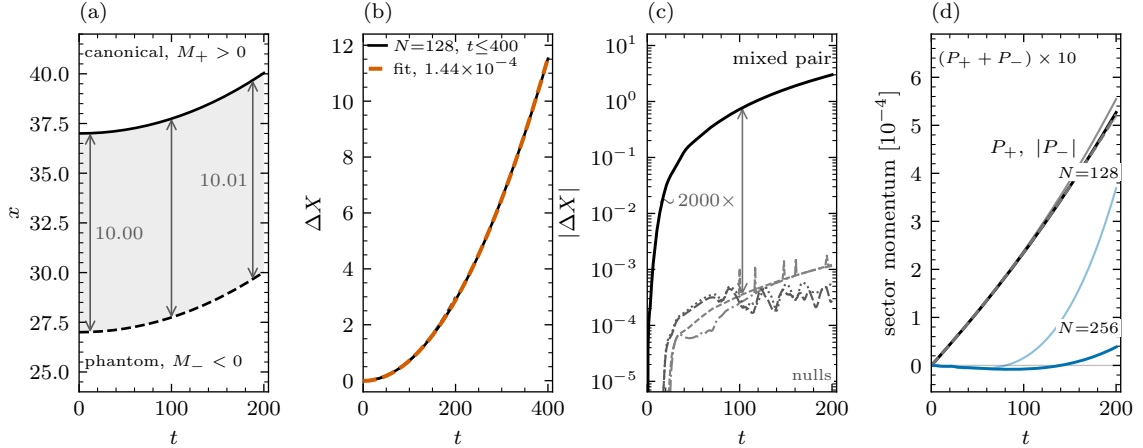


FIG. 4. (a) Core worldlines in the headline cell ($\Delta x = 0.25$): the phantom chases the canonical star it repels; the gap is marked at three times. (b) Midpoint drift of the $t = 400$ run (solid) with the single constant-acceleration fit over $5 \leq t \leq 400$ (orange dashed)—one parabola holds for the whole run—and the finest-grid $t \leq 200$ cell (dashed). (c) The signal against every null on a log scale: the mixed pair (barycentre midpoint) against the lone stars (core tracker) and the same-sign pair centroids—a separation of 3–4 orders of magnitude, sustained for the whole run. (d) The Bondi signature itself: each sector’s volume-integrated matter momentum climbs to $\sim 5 \times 10^{-4}$ while the *signed* sum of Eq. (4) (blue, magnified ten times) holds at zero—and holds better on the finer grid, the base rung being the fainter curve. Displacement without momentum, measured.

carrying no net momentum, exactly as the signed book-keeping demands. **Constant separation:** the barycentre gap ends at 9.915 and the core gap at 10.013, and the *proper* separation between the cores, $\int \sqrt{\gamma_{xx}} dx$ along the axis, runs $10.0001 \rightarrow 10.0244$ —so rigidity holds to 1% over $t \leq 200$ whether it is read in coordinates or in proper distance (the long run opens the gap late, and does so in proper distance too; Sec. VII).

The matter survives the acceleration. The field peak is flat to 0.3% over the run, the sector rms radii grow only from 4.3 to ~ 5.2 , the minimum lapse never leaves 0.991, and $\min \chi = 0.9889$ —nowhere near collapse ($\chi \rightarrow 0$) or dispersal. A lone star under the same numerics holds its peak to 0.5% for $t = 200$.

TABLE I. The falsifiable run matrix: thirty-four cells. The sign structure predicts a drifting midpoint for the mixed pair only, reversal under swapping the sectors, and a d^{-2} rate; every control row must return a null *centroid drift*—including the same-sign pairs, which merge into one lump and still do not displace their centroid. ΔX at $t = 200$ unless marked. The separation scan is $d_0 = 8, 12, 16, 20$; the mass ladders are the three retuned pairs at $d_0=10$ and the four equal-mass pairs at $d_0=20$; the stability scan covers $\omega = 0.75\text{--}0.90$. Full cell names and per-cell data are in the results pack (Sec. IX).

run	grid(s)	ΔX	verdict
PM, $d_0=10$	256^3	+3.0016	runaway
PM, $d_0=10$	192^3	+3.0139	converged pair
PM, $d_0=10$	128^3	+2.8815	4% low
PM, $d_0=8\text{--}20$	128^3	—	$a \propto d^{-2.03}$
MP mirror	128^3	−2.8815	reversed ^a
PM, deeper solve	128^3	+2.8815	+0.002%
PM, AMR on	128^3	+2.8815	+0.001%
PM, gauge $\eta=2$	128^3	+2.8813	−0.007%
PM, box doubled	256^3	+2.7606	−4.2%
PM, $t \leq 400$	128^3	+11.518	a steady
PM, $t \leq 784^d$	128^3	—	0.12c reached
PP (+, +) ^b	$128^3\text{--}256^3$	$\leq 7.8 \times 10^{-4}$	merges; null
MM (−, −) ^b	$128^3, 192^3$	$\leq 5.3 \times 10^{-4}$	merges; null
P, M lone stars ^c	128^3	$\leq 1.8 \times 10^{-3}$	null
M alone, $t \leq 1000$	128^3	—	−5.4% peak
mass ladders	128^3	—	$a \propto M^{0.97}$
stability scan	128^3	—	static

^a Drift ratio −1.000022, acceleration ratio −1.000010.

^b Both same-sign pairs coalesce into a single lump (Sec. VE); the entry is the largest pair-centroid excursion *through* that merger—the quantity the null tests. MM has two 128^3 cells; the second adds frame output.

^c Largest core displacement on any axis over $t = 200$; the phantom sits off-centre, the sharper test. Against the tracker the pair drift uses, the phantom’s own barycentre moves 1.3×10^{-2} (Sec. VE).

^d Two late-time follow-ups: a recentring box carried to $t = 784$ and a static-box control started off-centre to $t = 600$, which together show the gap opening is the grid and not the diagnostic (Sec. VII).

B. The force law: $a d^2$ returns the stars’ mass

Bondi’s law is quantitative: each star is accelerated by the *other’s* active mass, so $a = \bar{M}/d^2$. Five separations test it (Fig. 5a,b). A power law through the five accelerations gives $a \propto d^{-2.028 \pm 0.011}$ against -2 exact—and adding the widest point moved the exponent *toward* the exact value (four-point fit: -2.041). The normalization is the stars’ own mass: $a d^2/\bar{M}$ runs $1.031 \rightarrow 1.011 \rightarrow 1.003 \rightarrow 1.001 \rightarrow 1.002$. The close-pair excess is physical and behaves like the finite-size correction it must be—largest exactly where the two stars’ envelopes overlap most, decaying monotonically, gone by $d = 16$. The $d = 20$ cell rules out a measurement floor— $a d^2$ does not

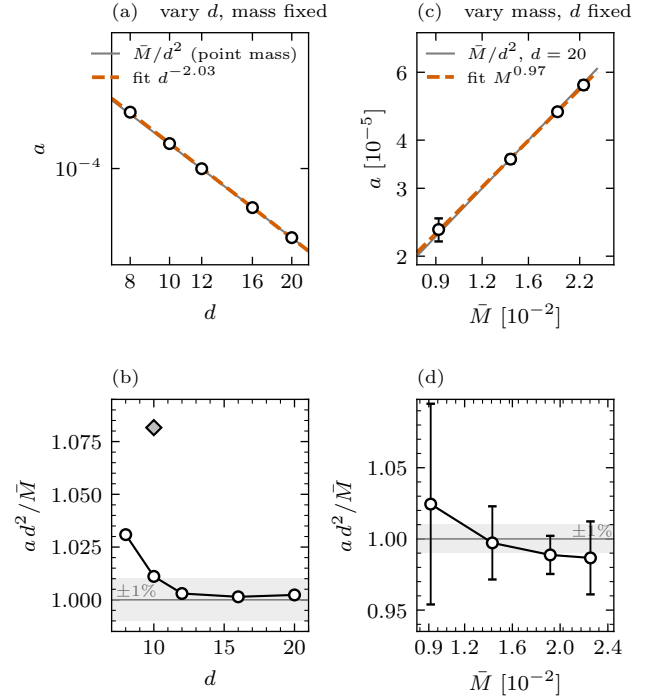


FIG. 5. Both axes of $a = \bar{M}/d^2$, measured. *Left, mass fixed:* (a) fitted acceleration against separation over $d_0 = 8\text{--}20$ at the base grid (open circles); the power-law fit (orange dashed) gives $d^{-2.03 \pm 0.01}$, and the point-mass line $\bar{M}/d^2$ (grey) is indistinguishable from it at this scale. (b) $a d^2$ over the pair mass $\bar{M}$: the close-pair excess decays monotonically, gone by $d = 16$. The open diamond is the converged ($N = 256$) value at $d = 10$; the $\sim 7\%$ base-grid calibration is common to the scan and cancels from the exponent and the trend (Sec. VC). *Right, separation fixed:* (c) the equal-mass ladder at $d_0 = 20$ —four pairs whose *two* stars are retuned together, spanning a factor 2.46 in mass—returns $a \propto \bar{M}^{0.97 \pm 0.06}$ against 1 exact. (d) The same ladder compensated: each rung recovers its own mass to within 2.4%, and the residual tilt carries the finite-size sign, the lightest pair being the most diffuse. Error bars are the half-spread of four disjoint-window fits. The ladder is fitted on the halo-free core tracker, which agrees with the whole-domain barycentre to 0.9% on every rung but the lightest, where the shed halo drags the barycentre fit 18% off.

plateau—and the signed residual momentum halves with each step outward too (5.8, 3.7, 2.3, 0.98, 0.49 in units of 10^{-5} at $t = 200$), tracking the acceleration rather than a diagnostic floor.

One systematic spans the whole scan: all five points share the base grid, whose acceleration at $d = 10$ sits $\sim 7\%$ below the converged fine-grid value (next subsection). The calibration is common mode and cancels from the exponent, from the trend in $a d^2$ and from every ratio in Sec. VF, but the *absolute* accelerations—2.307, 1.448, 0.998, 0.560, 0.359 in units of 10^{-4} —carry a grid uncertainty of that size: the converged $d = 10$ acceleration, $(1.55 \pm 0.01) \times 10^{-4}$, lands 8% above $\bar{M}/d^2$, so at a gap of two stellar diameters the point-mass formula holds to

ten percent rather than one. A converged separation scan is the natural sequel.

The law’s other axis is the mass. Three further pairs were built at $d_0 = 20$ —far enough out that the excess of Fig. 5b has decayed to 0.2%—with *both* stars retuned together, so each pair stays mass-matched and therefore rigid; an unequal pair deforms (Sec. VF), and a quadratic fit to a moving separation measures nothing. Equal masses need *different* frequencies in the two sectors (Fig. 2), so each phantom partner was root-found on its branch. With the archived $d_0 = 20$ cell as the fourth rung the ladder spans a factor 2.46 and returns $a \propto M^{0.966 \pm 0.061}$, 0.6σ from unity, each rung’s own $a d^2/\bar{M}$ landing within 2.4% of 1 (mean 0.999) with the residual tilt carrying the finite-size sign (Fig. 5c,d). Both axes of $a = \bar{M}/d^2$ are therefore direct measurements—and this one is compared rung by rung against its own prediction, not against a neighbour.

C. Convergence, and what cannot move the answer

Refining the grid does not erode the runaway—it settles it. The drift at $t = 200$ runs +2.881, +3.014, +3.002 down the ladder: the two fine rungs agree to 0.4% on drift and 0.5% on acceleration while the base rung sits 4% low, which is the opposite of what a discretization artefact does. **We quote $\Delta X(200) = 3.00 \pm 0.01$ and $a = (1.55 \pm 0.01) \times 10^{-4}$** , the fine-pair spread as the error bar. No formal convergence order accompanies this, for a measured reason (Fig. 6b): the $t = 0$ Hamiltonian norm *rises* with resolution ($2.0 \rightarrow 3.7 \rightarrow 5.6 \times 10^{-4}$, the transfer residual of Appendix A) while the evolution error falls ($1.0 \times 10^{-5} \rightarrow 5.4 \times 10^{-6}$ late-time), so the triple is non-monotone and Richardson extrapolation would return corrections smaller than the fine-pair spread at any assumed order.

Two twins of the base rung close the remaining loopholes with runs rather than arguments. *Is the base rung solve-limited?* A cell identical except for the elliptic tolerance—driven from $8.6 \times 10^{-4}\%$ to $1.9 \times 10^{-4}\%$, a $4.4\times$ deeper residual—changes the drift by 0.0015% and the acceleration by less than a part in 10^4 , while refining the grid moves it by 4%: the rung offset is a property of the evolution grid, not of the solve. *Does the uniform-grid choice shape the result?* The same cell with mesh refinement enabled reproduces drift and acceleration to 0.001%–0.002%—and the log shows level 1 was never created, because the tagger’s threshold ($|\chi - 1| = 0.02$) sits four times above anything these spacetimes produce. *Is the gauge shaping it?* Doubling the Gamma-driver damping ($\eta = 1 \rightarrow 2$), one parameter and nothing else, shrinks the shift by 5.0% everywhere while the drift moves by 0.007% and the acceleration by 0.019%: the coordinates changed by three hundred times more than the physics did. Finally, doubling the box (eight times the volume, sponge moved out) changes the drift by -4.2% : the boundary is not driving the motion.

D. The mirror: swap the sectors, the runaway reverses exactly

Swapping the two sectors in place must reverse the physics and nothing else. **It reverses to two parts in 10^5** : the mirrored cell returns drift ratio -1.000022 and acceleration ratio -1.000010 against the headline cell. What that excludes is precise: any asymmetry of the machinery that is *not* sourced by the matter—a biased stencil, a lopsided box, a directional diagnostic—since none of those knows which star carries the minus sign. It does not exclude a matter-sourced artefact, which reverses under the swap just as the physics does; the retired displaced-initial-data artefact of Sec. III B would have passed this test. Those are excluded instead by the force law, the mass ladder and the nulls that follow.

E. The nulls, and the two interaction channels

The lone-star rows calibrate the floor: a single star, alone in the box, moves at most 1.8×10^{-3} on any axis in $t = 200$ (canonical, box centre) and 1.6×10^{-3} (phantom, released *off-centre* at $x = 37$, where no symmetry protects it)—about $2000\times$ below the signal, with no preferred direction. Read against the tracker the pair’s drift uses, the like-for-like floor is the phantom’s own barycentre, which moves 1.3×10^{-2} over the same window—a factor 225, not the 1600 the core figure alone suggests; both are correct for what they measure, and the null rests on the smaller ratio.

The phantom was also carried alone to $t = 1000$, five times the standard window. It reproduces the $t = 200$ control digit for digit through their overlap, then keeps going without collapsing or dispersing—but it does not hold its amplitude indefinitely: the field peak decays *linearly* at 0.632% per 100 time units from $t \approx 200$ (disjoint sub-windows 0.604, 0.648, 0.608), reaching -5.4% at $t = 1000$. Everything else stays benign: $\min \chi$ rises $1.00000 \rightarrow 1.00033$ (a hill, not a well), the rms radius grows $4.41 \rightarrow 4.83$, the confined fraction falls $0.736 \rightarrow 0.670$, and the Hamiltonian norm falls $44\times$ as the initial transient radiates. The star relaxes outward and leaks slowly; it does not go unstable. The same curve reads -0.5% at $t = 200$ and -1.6% at $t = 400$, which is why no shorter run resolves it—and why the claim here is a measured decay rate over the window evolved, not an asymptotic one.

The same-sign pairs are the campaign’s sharpest control, and their behaviour was not anticipated: **they merge—both of them, on the same clock**. Tracked frame by frame (Fig. 7a), the two canonical stars of PP coalesce at $t = 33.6$ and the two phantoms of MM at $t = 32.8$ —a 2.4% difference, within the trackers’ grid disparity. That timing is the measurement. Newtonian gravity *attracts* the PP pair and *repels* the MM pair—Bondi’s own sign rules, Fig. 1—so if gravity set the timescale, PP would coalesce while MM flew apart. Instead both col-

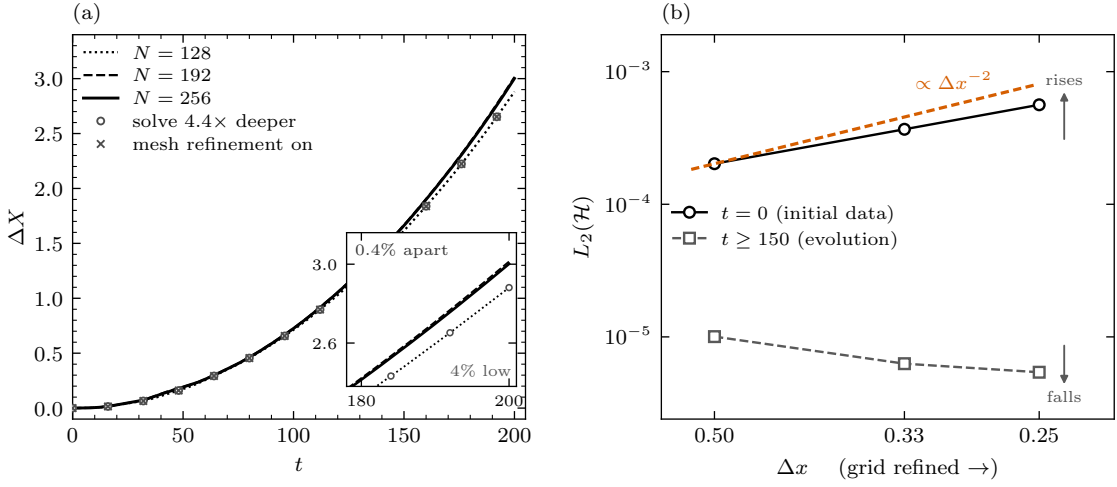


FIG. 6. (a) The resolution ladder at $d_0 = 10$: midpoint drift at $\Delta x = 0.50, 0.33, 0.25$, with two twins of the base rung overlaid—the elliptic solve driven 4.4× deeper (circles) and mesh refinement enabled (crosses); both reproduce it to 2×10^{-5} . Inset: the endpoint—the two fine grids agree to 0.4% while the base rung sits 4% low. (b) Two competing error sources: the $t = 0$ violation *rises* with resolution (the solve-to-evolution transfer residual, amplified as Δx^{-2} ; Appendix A) while the evolution’s own *falls*—two opposite grid behaviours never leave a single asymptotic power.

lapse identically: the driving force is blind to the sign of the mass, and its scale gives it away—Newtonian free fall from $d = 10$ at these masses takes $t \approx 207$, so the observed $t \approx 33$ needs a force $\sim 35\text{--}40\times$ gravity. That is the same-field overlap attraction of Sec. II, present exactly and only where two lumps share one complex field; Bondi’s — repulsion is real but invisible beneath it. The pairs are seeded in phase, the attracting configuration (Sec. III B), which is what allows them to fuse into a single centred lump.

This measurement is what certifies the mixed pair as gravitational. Its two stars are *different* fields with no coupling term, so the overlap channel is structurally absent, and the data confirm that absence three ways. At the same $d = 10$ where same-field pairs close in $t \approx 33$, the mixed gap holds to 1% for $t = 200$: a few-percent leak of a $35\times$ -gravity force across sectors would have closed it visibly. The mixed force is a clean $d^{-2.03}$ power law across $d = 8\text{--}20$, where a field-overlap force dies exponentially with distance. And its strength tracks the partner’s *ADM mass* (next subsection), a quantity a field force knows nothing about.

Meanwhile the same-sign cells deliver their null under the most violent history in the campaign: through plunge, merger, sevenfold ejecta growth and ringdown, their pair centroids never move more than 7.8×10^{-4} (PP) and 5.3×10^{-4} (MM) on any of six cells across three grids (Fig. 7b)—and the residual does not grow with resolution. Neither pair has a mass dipole, so neither centroid moves, *even during a merger*. The merger dynamics themselves are converged physics (ejecta growth $\times 7.1/7.0/6.9$ across the ladder), and the remnants carry the model’s structural sign difference: the PP remnant is a well in χ ($\min \chi = 0.979$), the MM remnant a *hill* (χ up to 1.011)—same matter behaviour, opposite curva-

ture. One diagnostic note: $\min \chi = 1.00000$ in MM does not mean flat geometry; a minimum only sees wells, and phantom curvature is all hills.

F. Gravity scales with the source, and the gap’s sign flips at equal mass

Each star is accelerated by the *other’s* mass, so retuning one star must change its partner’s acceleration and leave its own alone. Three one-knob variants test this (Fig. 8), retuning Φ_- to $\omega = 0.88$ and 0.804 and, for the phantom-heavier rung, the *canonical* star to $\omega = 0.81$. The constant C in $\ddot{x} = \pm C/d(t)^2$ is fitted per star by regressing its core’s $x(t)$ on $\{1, t, I_2(t)\}$ over $t \geq 5$, I_2 being the double time integral of the *measured* $d(t)^{-2}$, so the fit assumes no constant separation; only ratios are quoted, the constants absorbing the grid calibration of Sec. VB whose absolute counterpart the equal-mass ladder supplies. Mass factors 0.600, 0.803, 0.747 return retuned-partner pull ratios 0.618, 0.810, 0.756—within 0.9–3.1% across a factor 2.2—with unchanged-partner controls 1.028, 1.011, 1.001. No rung exists below 0.60 of the matched mass: both branches floor at $|M|_{\min} \simeq 0.54$ of it (Fig. 2), which is itself a result. And the ladder carries a signature with no artefact analogue: **the gap’s behaviour flips sign exactly where the masses cross**. Heavier phantom: the gap opens (+0.60). Matched: it holds (+0.003). Lighter phantom: it closes (−0.59, −1.38), monotonically. Interpolating between the two cells that bracket the flip puts it at mass ratio 0.995 ± 0.03 , against 1.000 predicted; the error bar is the spread over end-of-run gap conventions (barycentre 0.995, proper 0.994, core 0.961, late-window mean 1.002), which dominates the statistical one. No grid, boundary or solver

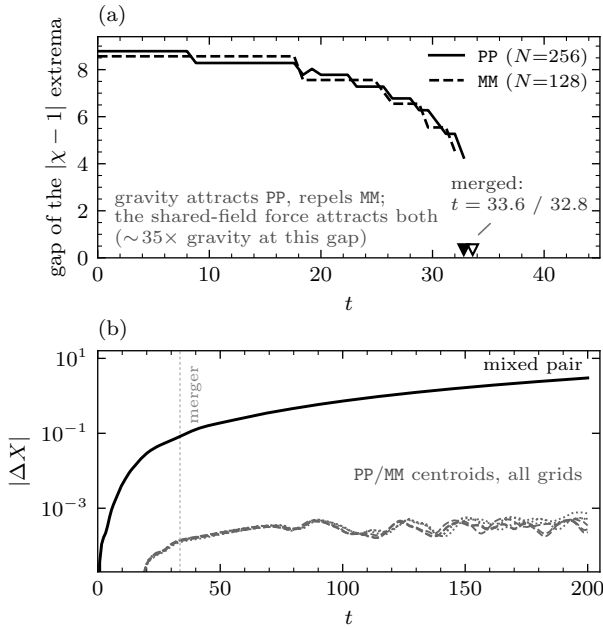


FIG. 7. The same-sign controls. (a) Gap between the two $|\chi - 1|$ extrema—wells for PP, hills for MM—tracked frame by frame: *both* pairs fall together and merge, at $t = 33.6$ and 32.8 . Newtonian gravity’s sign is *opposite* between these two cells, so their common clock is set by the sign-blind shared-field attraction, $\sim 35\times$ gravity at this gap—a channel the mixed pair structurally lacks. (b) Through plunge, merger and ringdown, the PP/MM pair centroids (all grids) stay below 8×10^{-4} —four orders under the mixed pair’s drift. A configuration that cannot manufacture momentum even while merging is a stronger null than one that sits still.

distinguishes which of the two stars is heavier; gravity does.

VI. LIMITS ON GRAVITATIONAL RADIATION AND NEAR-ZONE BEHAVIOUR

A uniquely relativistic aspect of the runaway is its radiation: the exact accelerating-pair metrics are radiative [2, 3], and negative-mass binaries have been proposed as sources of anomalous signatures [39], with the leading anomalous channel for mismatched charge-to-inertia binaries being dipolar. The bicomplex model closes that channel structurally: the source of gravity is the signed stress tensor, conserved on shell (Sec. II), so the rate of change of the signed mass dipole is the conserved signed momentum—zero for this data, and measured to cancel to $\lesssim 1\%$ (Sec. VA)—and mass-dipole radiation is forbidden at leading order for exactly the reason it is forbidden in ordinary general relativity. Kinematics agrees from the other side: Ψ_4 carries spin weight -2 , so its expansion begins at $\ell = 2$ whatever the dipole does. What survives is quadrupolar bremsstrahlung from the accelerating pair—secular, chirpless, with no astrophysical counterpart, and for this configuration very small.

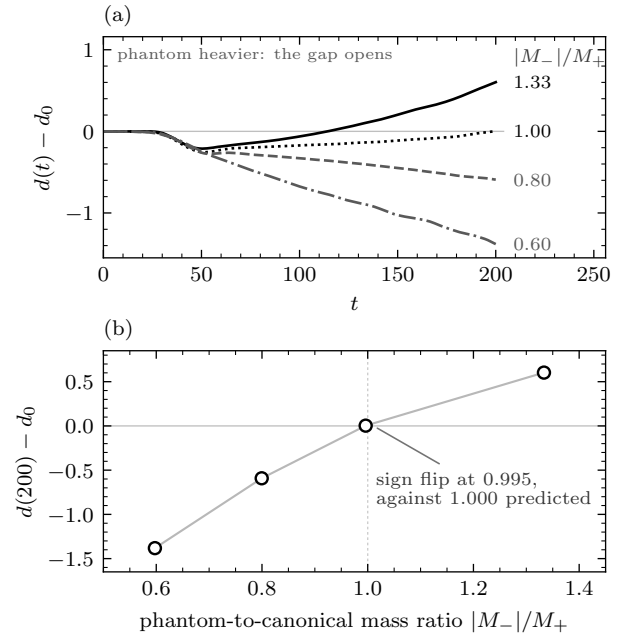


FIG. 8. The mass ladder: the matched pair and three retuned partners spanning a factor 2.2 in mass ratio. (a) Gap history: with a heavier phantom the gap *opens*, matched it holds, with a lighter phantom it *closes*, monotonically in the ratio. Every cell shares the same $t \approx 50$ gauge dip, so the gap is read late. (b) Gap change at $t = 200$ against the pair mass ratio $|M_-|/M_+ = 0.597, 0.800, 0.996, 1.333$, monotone through the four cells: the sign flip falls at ratio 0.995, against 1.000 predicted.

The signed mass quadrupole of a zero-momentum pair is $Q \simeq 2\bar{M}dX(t)$, so at constant separation and constant acceleration $\ddot{Q} = 2\bar{M}ad$ is *constant*: measured at 4.5×10^{-5} against 4.4×10^{-5} predicted, steady to 7% over $20 \leq t \leq 190$. A constant $\ddot{Q}$ radiates nothing at leading order, and the flux enters only through the slow secular evolution of d and v —from the trajectories $|\dot{Q}| \leq 6 \times 10^{-8}$, a quadrupole luminosity $\sim 4 \times 10^{-17}$ and $\sim 7 \times 10^{-15}$ radiated over the run, ten orders of magnitude below the pair’s total mass of 5.5×10^{-5} . So the $\ell = 2$ content is bounded rather than measured here, and the bound is far above what the model expects: every extraction shell the domain affords lies inside the near zone, so what follows is an upper limit on the amplitude.

The doubled box measures how much of that reaches the wave zone at these speeds: **none that we can detect out to $R = 40$** (Fig. 9). The $\ell = 2$ amplitude falls as $r^{-4.8}$ across shells spanning $R = 16\text{--}40$ — $r\psi_4$ drops by a factor 31 where one outgoing wave would keep it flat—so every shell sits in the near zone, reading the source’s Coulombic curvature rather than a flux. We report this as a null with a number: it bounds the radiated content of the run, agrees with the structural argument above, and defines the follow-up—evolve the pair to relativistic speed (only 0.03c inside the fit window here), where bremsstrahlung scales steeply and the near-zero

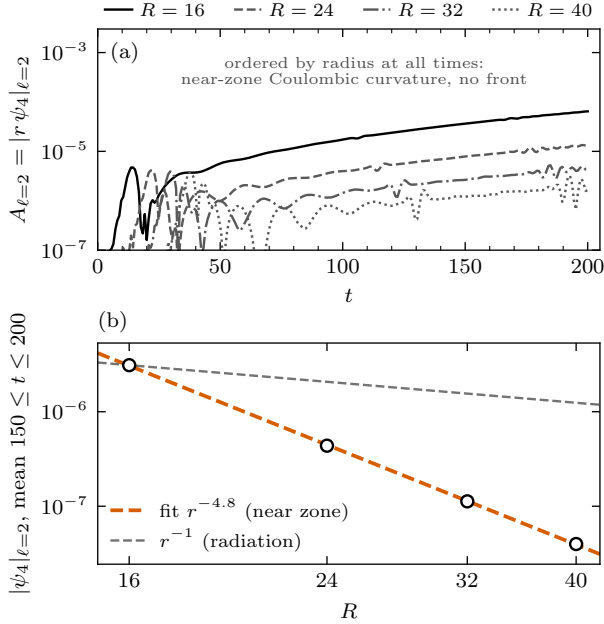


FIG. 9. The wave-zone measurement on the doubled box ($L = 128$, $\Delta x = 0.25$, sponge at 48–60). (a) Total $\ell = 2$ amplitude $A_{\ell=2}$ on the four extraction shells: ordered by radius at all times, with no propagating front. (b) The falloff, averaged over $150 \leq t \leq 200$: $|\psi_4|_{\ell=2} \propto r^{-4.8}$, against r^{-1} for radiation. Every shell out to $R = 40$ is in the near zone, so $A_{\ell=2}$ is a Coulombic near-field amplitude, not a radiated flux—consistent with the quadrupole estimate of Sec. VI, ten orders of magnitude below this bound.

total mass budget becomes measurable (Sec. VIII).

VII. SYSTEMATICS, WITH THEIR MEASURED SIZES

Fit convention. Fitting from $t \geq 5$ against fitting the last two thirds moves any single a by 1–3% (a late window weights the fastest part of the run); every number here uses $t \geq 5$, and the *trends*—the exponent, the ratios, the sign flip—are insensitive to the choice.

Grid calibration. The base grid underestimates the acceleration by $\sim 7\%$ at $d_0 = 10$; the fine pair agree to 0.5% and set the quoted values. The offset is common to the base-grid scans and cancels from their differential statements (Sec. VB).

Rigidity, and what opens the gap. Constant separation is a 1%-level statement on $t \leq 200$: the two trackers bracket the gap change between -0.9% and $+0.1\%$. Past that the gap opens—11.8 by $t = 400$, 18.2 by $t = 600$ —while the acceleration stays steady, so “rigid pair” belongs to $t \leq 200$ and “constant rate” does not depend on it. Two follow-up cells identify what opens it. Not the diagnostic box: a recentring box carried to $t = 784$ and a static-box control started ten units off centre agree on the separation to 1.5% over their whole

600-unit overlap, crossing rather than diverging, their sector field contents agreeing to 0.3%. The grid. At $t = 200$ the opening runs $+0.107$, $+0.029$, $+0.013$ down the resolution ladder—an 8.4-fold reduction over a factor two in spacing, third order ($p = 3.08$)—and a physical separation does not converge away under refinement; its sign and size match the ~ 0.1 -cell initial-data offset of Sec. IIIB, which pushes the phantom out of its own well while the canonical star falls into it. The $t \leq 200$ result is unaffected, being quoted at the converged rung where the opening is 0.13%; what is new is that the base-grid error grows fast enough to dominate by $t = 400$. In those cells the pair passes $0.10c$ at $t = 600$ and $0.12c$ at $t = 784$, its acceleration peaking near $t = 400$ and down a third by the end—so relativistic speed needs a converged grid, not a longer run: the pair stops being a pair first.

Coordinates and diagnostics. Drift and gap are coordinate quantities in a dynamical gauge, and the whole-domain barycentre of a weakly gravitating sector is sensitive to domain noise: that of the off-centre lone phantom wobbles by 4×10^{-2} early (domain noise entering the sector weight) while its core never leaves 1.6×10^{-3} —which is why nulls are quoted on cores and the pair on both trackers, agreeing to 1%. The gauge is also checked directly rather than argued around. The shift at the two cores never exceeds 7.7×10^{-4} , and it is very nearly *antisymmetric*— β^x points one way at the canonical star and the other at the phantom—so the part of it that could translate the pair, the mean of the two, stays below 1.1×10^{-5} against a measured midpoint velocity of 3.0×10^{-2} : three orders of magnitude apart. Integrated over the run, that mean shift accounts for 4.0×10^{-4} of coordinate displacement, 0.013% of the measured $\Delta X = 3.00$. The drift is therefore matter translating through the Eulerian frame—anchored at the asymptotically flat Sommerfeld boundary—not coordinates dragged along with it; the contrast is with moving-puncture evolutions, where β^i tracks the punctures by construction. The proper separation of Sec. VA puts rigidity on the same semi-invariant footing, and the $\eta = 2$ twin of Sec. VC closes the question by run as well as by measurement. The measurements that do not share the coordinate choice at all—the mirror ratio, the signed-momentum cancellation, the same-sign merger clock, the χ hill/well structure—corroborate the coordinate ones. For the record: the zero initial shift is inconsistent with the outer boundary condition and emits a flat $\sim 10^{-6}$ gauge pulse inward from $t = 0$; it crosses the stars before $t = 20$ and leaves no trace in the constraints or the fitted acceleration.

Same-sign initial data. The conformally flat outer boundary condition of the elliptic solve is genuinely wrong for a box carrying net mass $2M$, so the same-sign solves floor near $10^{-3}\%$ rather than converging arbitrarily; this is a $t = 0$ statement only—their evolution constraints stay bounded through the merger—and it is one more reason those cells serve as momentum nulls rather than precision measurements.

VIII. DISCUSSION AND CONCLUSION

We posed Bondi’s positive–negative mass pair as a full initial-value problem, and it behaved as he argued in 1957—quantitatively.

(1) **The runaway is real, and it is steady.** Released at rest, the mixed pair self-accelerates at constant rate for as long as we evolve it, reaching $0.056c$ by $t = 400$ with the rate flat to 2%; swapping the sectors reverses the drift to two parts in 10^5 ; the separation holds to 1%, in proper distance as in coordinates.

(2) **It is gravity.** By construction the sectors meet only in the metric; by measurement the force falls as $d^{-2.03 \pm 0.01}$ with ad^2 returning the stars’ mass, and on a mass-matched ladder spanning a factor 2.46 the mass exponent itself comes back as 0.97 ± 0.06 ; differentially, the pull tracks the partner’s ADM mass through a factor 2.2—including the gap’s sign flip at equal mass. The same-sign controls isolate the model’s only other channel, the shared-field overlap attraction, and show it $\sim 35\times$ gravity, short-range and sign-blind—present where the model puts it, absent from the mixed pair.

(3) **Displacement without momentum—the Bondi signature.** The sector momenta, each growing to 5×10^{-4} , cancel in the signed sum to under 1%, the residual converging *toward* zero with resolution while the displacement converges onto 3.00 ± 0.01 . Same-sign pairs, having no mass dipole, hold their centroids at the 10^{-4} level *even while merging*.

(4) **A star of negative ADM mass exists** as constraint-satisfying, asymptotically flat initial data, and survives 400 time units of evolution breathing at the sub-percent level (static phantom-field stars [31] and de Sitter bubbles [29] were known; a negative-mass *horizon* is a different object [12]). It is the better-behaved of the pair: shed field is pushed away rather than re-accreted, so the loop that lets canonical stars deepen their own wells runs backwards. Alone and carried to $t = 1000$ it neither collapses nor disperses; it relaxes outward, its central amplitude falling linearly at 0.63% per 100 time units, so what is established is a measured decay rate over the window evolved, not asymptotic stability.

What is not established. The continuum force-law normalization away from $d = 10$ (the scan is base-grid; Sec. VII); rigidity beyond $t = 200$, where the gap opens while the rate holds—attributing the opening to discretization error is an inference from its sign, size and third-order scaling, and a late-time rung at $N = 192$ would settle it; transverse stability, every release here being head-on and axisymmetric about the pair axis; and the radiation. On the last, the near-zone null out to $R = 40$ agrees with both the structural no-dipole argument and the ten-orders-below quadrupole estimate of Sec. VI, but the energy budget of a radiating system whose total mass starts at 5.5×10^{-5} —0.4% of either star, with no positive-mass theorem to floor its descent—is the open question this configuration was built to reach. It needs speed, and speed is now a resolution problem

rather than a wall-clock one: a recentring-box chase reaches $0.12c$ at the base grid before the pair’s own discretization error opens the gap and stalls the acceleration, so $\sim 0.3c$ —where the anomalous-signature programme of Ref. [39] meets a concrete source—is a converged-grid measurement waiting to be made. Nothing here bears on whether such matter exists.

IX. REPRODUCIBILITY

Every number and figure derives from the packed campaign at <https://github.com/Nikchik-coder/GRTEclyn/tree/develop/results/bondi-dipole-runaway> [44]: one directory per cell under `campaign/`, each holding the tracker streams, constraint norms, Weyl modes, solver history, and the exact launch environment. Cell names carry the configuration (`runaway_pair_d10_L64_N256_lev0`: the mixed pair at $d_0 = 10$, $L = 64$, 256^3 , uniform), and `stars/` holds the dressed-star profile tables and both family scans. A single script (`article_figures.py`, in the repository’s visualisation package) regenerates every figure and prints every quoted number, the gauge, energy-condition and radiation-budget diagnostics of Secs. VI and VII included. The evolution and initial-data code lives in the same repository [44]; the version behind this paper is identified by its git tag, and an archived snapshot carrying a DOI accompanies the published version. Movies of the headline cell and of both same-sign controls are submitted as Supplemental Material. Initial data: GRTresna [25]; evolution: GRTEclyn [23, 24, 26].

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Appendix A: Constraint behaviour

Initial data. Every mixed-pair solve converges monotonically on the maximal-slicing path and exits under its gate: $8.6 \times 10^{-4}\%$ (Hamiltonian) and $8.0 \times 10^{-4}\%$ (momentum) at the base grid in 13 nonlinear iterations, tightened with the grid to $8.7 \times 10^{-5}\%/5.1 \times 10^{-5}\%$ on the finest rung (16 iterations)—so the fine rungs do not inherit the coarse rung’s residual, unlike a fixed-tolerance ladder. The deep-solve twin of Sec. V C shows the evolution is insensitive to a further $4.4\times$ of solve depth at

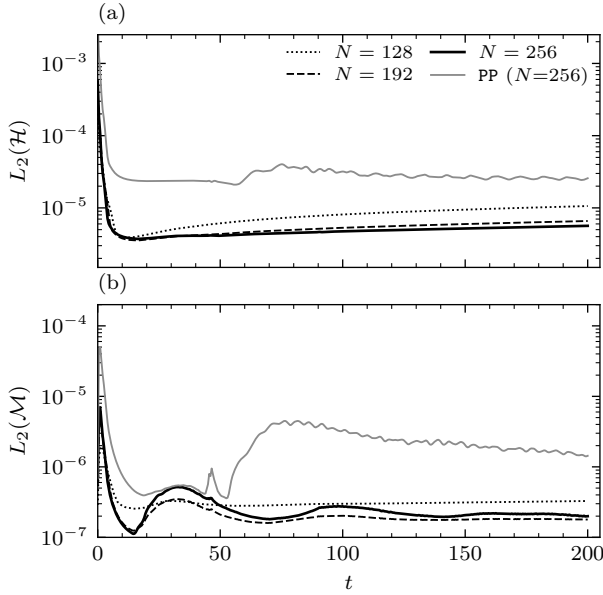


FIG. 10. Constraint norms for the headline cell’s resolution ladder (line styles as in Fig. 6) and the merging PP control at the finest grid (gray). (a) Hamiltonian, L_2 over the domain: finer grids start worse (initial-data interpolation noise) and evolve better. (b) Momentum, L_2 .

fixed grid, because the $t = 0$ violation on the *evolution* grid is dominated by the solve-to-evolution transfer, which scales as Δx^{-2} and which no solver tolerance can touch: the deeper solve lowered it by only $\sim 1\%$ ($2.02 \rightarrow 2.00 \times 10^{-4}$). The mechanism is the order of the handoff of Sec. III B: the metric–matter mismatch is a grid-scale quantity whose amplitude does not fall with the grid, while the operator reading it is a second difference carrying Δx^{-2} ; removing it means matching the two discretizations at the order of the constraint operator—machinery to be built in its own right. Its absence is bounded instead of ignored: the spike is a $t = 0$ object that radiates away, falling by an order of magnitude within $t \lesssim 3$ ($t = 3.0, 2.0, 1.5$ down the ladder—faster on finer grids, as a grid-scale transient should) and settling onto its late plateau by $t \approx 4$, before the $t \geq 5$ fit window that every number in this paper uses opens.

Evolution. Figure 10 shows the headline ladder in ab-

solute code units (the domain is overwhelmingly vacuum, so relative norms are dominated by near-zero denominators; the yardstick is the controls under identical numerics). Each rung relaxes from its initial-data spike within $t \lesssim 2$ and then holds nearly flat for the whole run: late-time $L_2(\mathcal{H})$ of 1.0×10^{-5} , 6.3×10^{-6} , 5.4×10^{-6} down the ladder—falling with resolution while the $t = 0$ spike rises, the two-error-source structure of Fig. 6b. The PP control runs a factor few above the mixed cell after its merger and stays bounded to $t = 200$; its centroid null holds through that entire history.

TABLE II. Measured evolution cost per cell type (one H100 per run, $t = 200$ unless noted; “frames” adds the movie/still pipeline). Device memory is $\sim 6/20/48$ GB at $N = 128/192/256$, which is what allows uniform 256^3 on one card where a refined grid would not fit.

cell type	grid	Δx	wall
pair, $L = 64$	128^3	0.50	1.1 h
pair, $L = 64$	192^3	0.33	4.8–6.0 h
pair, $L = 64$, frames	256^3	0.25	14.6 h
wave zone, $L = 128$	256^3	0.50	7.4 h
long run, $t = 400$, frames	128^3	0.50	3.8 h
lone star, $t = 1000$	128^3	0.50	5.4 h
pair, $t = 784$, moving box	128^3	0.50	4.3 h

Appendix B: Computational cost

Every evolution ran on NVIDIA H100 80 GB devices (one per run) on a node carrying four of them behind two 32-core Intel Xeon Platinum 8462Y+ processors; the elliptic solves ran CPU-side on 32 MPI ranks, ~ 20 minutes at 256^3 and ~ 4 hours at 512^3 . GRTeclyn was built against AMReX with CUDA 12.1 and GCC 11.4; the commit each cell was built from is recorded in its metadata. At $\Delta t = 0.02 \Delta x$, a $t = 200$ evolution takes 20 000/30 000/40 000 steps at $N = 128/192/256$; measured wall times are in Table II. The thirty-four packed cells total $\simeq 99$ GPU-hours of evolution; a two-GPU evolution path was smoke-tested and works at these sizes, but four single-GPU cells in flight is the better allocation of the node.

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